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CHAPTER I

INTRODUCTION

1.1 Background of the Study

As the retail and wholesale distribution business continues to grow, the need for efficient ways to manage inventory has become increasingly important. Cleaning supply distributors take care of products such as detergents, disinfectants, mops and brooms, and sanitation materials that are redistributed to retailers as well as offices, schools, hospitals, and other institutions. This includes effective management of stock to provide timely availability of products, minimize losses, and keep customers satisfied.

Inventory management plays a vital role in the daily operations of a distribution business. It involves monitoring stock levels, tracking product movement, recording stock-in and stock-out transactions, and ensuring that products are available whenever customers need them. Proper inventory control supports operational efficiency and decision-making and is consistent with established operations-management principles [1].

Many small and medium-sized cleaning supply distributors still rely on notebooks, logbooks, or spreadsheet files to monitor inventory. These practices can result in inaccurate stock counts, lost records, duplicated entries, delayed reporting, and difficulty monitoring fast-moving and slow-moving products. Real-time inventory visibility and updated information can support replenishment and supply-chain decisions [2].

The proposed Intelligent Stock and Sales Analysis System addresses these concerns by providing a centralized platform for product records, stock transactions, sales transactions, reports, and sales analysis. The system is intended to reduce manual work and make current information easier to retrieve.

1.2 Statement of the Problem

- Inventory is recorded manually, resulting in inaccurate stock monitoring.
- Stock-in and stock-out transactions are not tracked in real time.
- Fast-moving and slow-moving products are difficult to identify.
- Restocking decisions are not sufficiently supported by summarized sales information.
- Inventory and sales reports are time-consuming to prepare and may contain errors.
- Inventory records are not centralized and are difficult to access quickly.
- Low-stock and reorder conditions are not automatically identified.
- Encoding errors can create discrepancies between physical and recorded stock.
- Returned, defective, or expired products are difficult to track.
- Historical sales information is not readily available for demand-related analysis.

1.3.1 General Objective

The general objective of this study is to design and develop an intelligent inventory and sales analysis system for a cleaning supply distribution business to improve stock control, reporting accuracy, operational efficiency, and decision-making processes. The system aims to automate inventory monitoring, sales tracking, and report generation to help the business

manage products more effectively and reduce manual work. Furthermore, the study seeks to provide a reliable and organized platform that supports real-time inventory management, minimizes errors, and enhances overall business performance.

- The proposed system will serve as a comprehensive technological solution designed to address the most common inventory problems experienced by the business, such as inaccurate stock records, delayed reporting, overstocking, stock shortages, and inefficient monitoring of fast moving and slow-moving products. By centralizing all inventory-related data into a single, integrated platform, the system will minimize manual encoding errors and discrepancies between actual and recorded stock levels. This will lead to more reliable inventory records, which are essential for day-to-day operations, financial reporting, and overall business planning.To reduce delays in reporting, the system will generate real-time updates on stock movements, including purchases, sales, returns, and adjustments. Automated report generation will allow management and staff to quickly access up-to-date information on inventory status without having to manually compile data from multiple sources. This timely access to information will improve operational efficiency and support proactive decision-making, rather than relying on outdated or incomplete reports.

- In addition, the study intends to improve the accuracy and accessibility of inventory information by maintaining a centralized database that can be updated in real time a fundamental upgrade from the fragmented, manual, or spreadsheet-based systems commonly used in small and medium-sized enterprises. In traditional setups, inventory data is often recorded in separate logs, notebooks, or local files, leading to inconsistencies, duplication, or outdated records that make it difficult to know exactly what stock is available, where it is located, or how much has been sold. This fragmentation creates gaps in information, leading to errors such as recording the same item twice, failing to update quantities after a sale, or having different departments rely on conflicting figures. By establishing a single, unified database, all inventory-related data including product details, quantities received, quantities issued, current stock levels, supplier information, and storage locations is stored in one secure location.

- Every transaction, whether it involves receiving new supplies, moving items between storage areas, or fulfilling customer orders, is recorded instantly and reflected across the entire system. This real-time updating ensures that the information available is always current, precise, and consistent, eliminating discrepancies and providing a single source of truth that every authorized employee can rely on. Whether accessed by warehouse staff, sales personnel, or management, the data remains identical and accurate, removing confusion and ensuring that every decision or action is based on the same reliable facts.

- The development and implementation of the intelligent inventory and sales analysis system represent a strategic initiative designed to modernize and strengthen the core internal operations of the business. At its core, this project aims to address longstanding operational challenges rooted in traditional workflows specifically the heavy reliance on paper-based processes and manual calculations, which have historically been time-consuming, prone to errors, and difficult to manage. By transitioning to a fully digital, automated, and intelligent system, the business expects to achieve significant improvements in efficiency, accuracy, data security, and overall operational agility, creating a more resilient and competitive foundation for daily activities and long-term growth.One of the most impact benefits of the system is the reduction and eventual elimination of dependence on paper-based documentation and manual computation. In current operations, tasks such as recording stock movements, updating inventory levels, tracking sales transactions, and generating performance reports require extensive paperwork and handwritten or spreadsheet-based calculations. These methods are not only labor-intensive but also carry high risks: data entry mistakes, misplaced documents, damaged records, and inconsistent recording practices often lead to discrepancies between actual stock and recorded figures, delayed information sharing, and flawed decision-making. The new system will digitize every step of these processes, automating data capture, calculation, and record-keeping. This shift will cut down the time spent on repetitive administrative work, minimize human error, and ensure that all operational activities follow a standardized, consistent workflow across all departments and teams.

- Moreover, the study aims to introduce data-driven decision-making practices within the business a strategic shift that moves operations away from guesswork, intuition, or past habits toward choices grounded in factual, timely, and measurable information. In many small and medium sized enterprises, decisions regarding how much stock to

order, which products to prioritize, or how to respond to sales trends are often made based on rough estimates or personal experience, which can lead to costly mistakes such as overstocking, under-stocking, or missed opportunities. By integrating modern information technology, this research seeks to change that culture by making data the central foundation of every operational and strategic choice.Through the intelligent system, the business will have access to automatically generated reports, clear graphical analyses, and detailed inventory statistics that are updated in real time. These tools transform raw numbers into understandable and actionable insights, eliminating the need for manual compilation or complex calculations that are prone to error. For instance, managers can view visual charts showing sales performance over weeks, months, or seasons, compare the movement of different cleaning supply items, and track how demand shifts in response to market changes, promotions, or seasonal needs. Inventory statistics will highlight key metrics such as stock turnover rates, average holding periods, reorder points, and the proportion of active versus idle products information that is essential for effective planning.

- Ultimately, this study seeks to demonstrate how the use of modern information technology can improve inventory and sales management in small and medium-sized enterprises (SMEs), which form the backbone of local economies yet often face unique operational challenges. Unlike large corporations that can afford complex enterprise resource planning (ERP) systems, many SMEs still rely on manual methods, paper records, or basic spreadsheets to track stock levels and sales transactions. While these traditional approaches may work for very small operations, they become increasingly inefficient, error-prone, and difficult to manage as the business grows, product lines expand, and customer demand becomes more dynamic. Common issues include inaccurate stock counts, delayed recording of transactions, difficulty in tracking product movement, lack of real-time data, and an inability to generate meaningful insights for planning all of which can lead to overstocking, stockouts, wasted resources, lost sales opportunities, and reduced profitability.Against this backdrop, the intelligent inventory and sales analysis system developed and examined in this research is designed specifically to address these pain points, tailored to the needs and scale of businesses dealing in cleaning supply products. The cleaning supply sector has distinct characteristics: it includes a wide variety of items ranging from daily consumables like detergents, disinfectants, and paper products to specialized equipment and chemicals; products often have varying shelf lives, storage requirements, and usage rates; and demand can fluctuate based on seasonal trends, commercial contracts, or sudden market shifts. Managing such a diverse range of items manually is not only labor-intensive but also carries high risks of mismanagement for example, holding too much stock of slow-moving items ties up capital and storage space, while running out of high-demand essentials can drive customers to competitors.

1.3.2 Specific Objectives

Specifically, the study aimed to:

1. Analyze the current inventory process of the cleaning supply distributor.
2. Design a user-friendly inventory and sales management interface.
3. Develop a system for recording stock-in and stock-out transactions.
4. Implement real-time inventory monitoring.
5. Integrate sales analysis features to identify product movement trends.
6. Generate automated inventory and sales reports.
7. Test and evaluate the system’s functionality, usability, and performance. Design a user-friendly inventory and sales management interface.

1.4 Scope and Limitations

The proposed system focuses on a small-scale cleaning supply distribution business. Its scope includes product management, stock monitoring, stock-in and stock-out recording, sales recording, reporting, dashboard information, and basic sales analysis. The system is intended to centralize operational records in a database and provide information that can be used for inventory and sales decisions.

The documented limitations include the absence of bar-code hardware integration, a dedicated mobile application, biometric authentication, Internet of Things-based monitoring, and full accounting functions such as payroll, automatic tax computation, general-ledger maintenance, and formal financial auditing. Evaluation is limited to selected participants and feasible academic testing conditions.

1.5 Significance of the Study

The study is significant to the business owner because it provides centralized inventory and sales information for monitoring and decision-making. It is also intended to assist staff in recording stock transactions and checking current product information. Customers may benefit indirectly from improved product availability and more organized service. Future researchers may use the project as a reference for related inventory, sales analysis, and information-system studies.In this industry, where companies handle a wide variety of products ranging from fast-moving consumables, cleaning chemicals, and disposable items to bulk supplies and specialized equipment effective inventory management is not just a support activity but one of the most essential operational functions. It forms the backbone of the entire distribution process, directly influencing critical business outcomes such as product availability, customer satisfaction, operational efficiency, and overall profitability. When inventory processes are poorly managed or rely on outdated methods, the business faces risks such as frequent stock shortages, overstocking of slow-moving items, product spoilage (especially for goods with shelf-life requirements), lost sales opportunities, and unnecessary holding costs, all of which negatively impact performance and revenue.

1.6 Conceptual Framework

The conceptual framework of the study is presented in **Figure 1.1**. It follows the Input–Process–Output (IPO) model, which shows how the problems and requirements of the cleaning supply distribution business are transformed, through a structured development process, into the Intelligent Stock and Sales Analysis System.

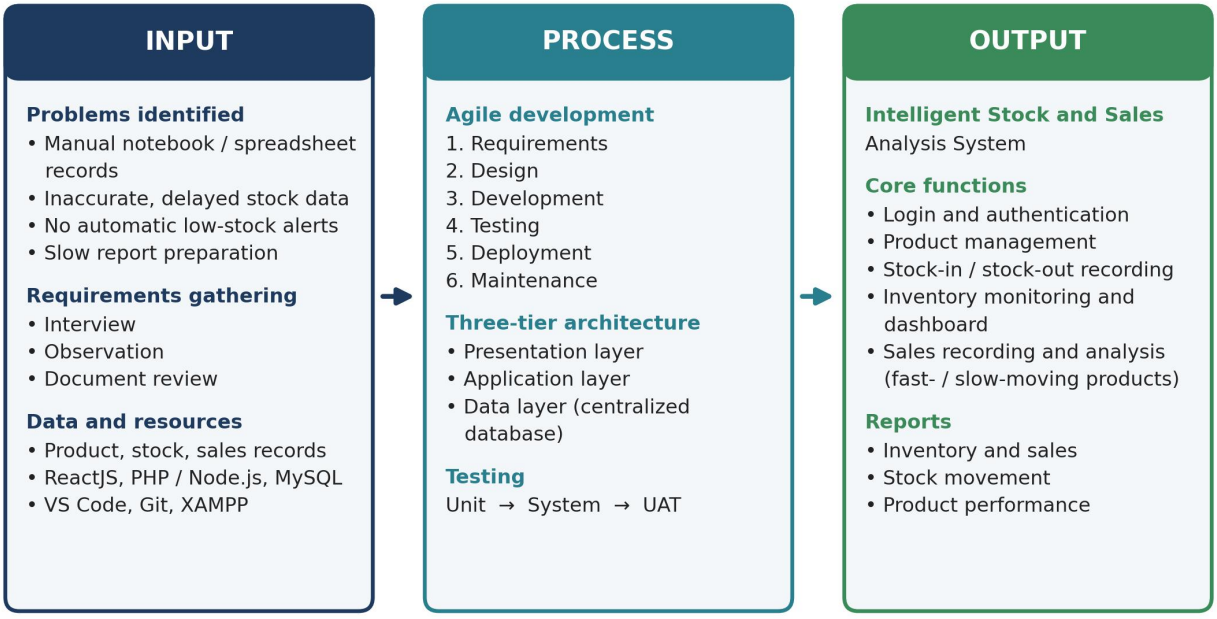


Figure 1.1. Conceptual framework of the Intelligent Stock and Sales Analysis System.

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CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Foreign Studies

Stevenson (2020) discussed inventory management as an important part of operations management, particularly in controlling stock levels, forecasting demand, and managing the movement of goods. The study emphasizes the importance of maintaining accurate inventory information to support operational decisions and avoid problems associated with excessive or insufficient stock. This is relevant to the present study because the proposed Intelligent Stock and Sales Analysis System for Cleaning Distribution focuses on improving stock monitoring and maintaining accurate inventory records. Similar to Stevenson's discussion, the proposed system provides centralized product information and stock transaction records. However, the present study applies these inventory-management concepts specifically to a cleaning supply distribution business, where products such as detergents, disinfectants, mops, brooms, and sanitation materials must be monitored regularly. [1]

Chopra and Meindl (2019) emphasized the importance of information visibility in supply-chain management. Their work explains that updated information about inventory and demand can help organizations make better replenishment decisions and respond to changes in customer requirements. This study is directly relevant to the proposed system because one of its objectives is to provide updated inventory information and support stock monitoring. The proposed system applies this concept by recording stock-in and stock-out transactions and providing sales information that can be used to identify products that require replenishment. While Chopra and Meindl discuss supply-chain management in a broader context, the present study focuses on applying inventory visibility to a small-scale cleaning supply distributor. [2]

Gupta and Kohli (2021) discussed the use of sales analytics for understanding product demand and identifying fast-moving and slow-moving products. Their work highlights how sales information can be analyzed to support inventory forecasting and purchasing decisions. This is relevant to the present study because sales analysis is one of the main features of the proposed system. The system uses recorded sales information to identify product movement trends and provide information that can assist in inventory planning. Unlike a general sales analytics application, the proposed study combines sales analysis with inventory monitoring so that product movement can be considered together with current stock information. [3]

Heizer, Render, and Munson (2020) discussed inventory, forecasting, and supply-chain management as important areas of operations management. Their work emphasizes the need to maintain appropriate inventory levels to avoid both excessive stock and shortages. This supports the present study because the proposed system provides inventory monitoring, sales information, and reporting functions that can assist the business in managing product availability. The study applies these operations-management concepts within the specific environment of a cleaning supply distributor. [4]

Heizer and Render explained that operations management involves coordinating resources efficiently to achieve organizational goals. Inventory systems play a critical role in this process because inventory represents a major business asset. Poor inventory management can lead to financial losses, operational delays, and reduced customer satisfaction.

Slack, Brandon-Jones, and Johnston (2020) conducted a study that explored the direct link between effective inventory management and overall operational efficiency. Their research revealed that businesses which adopted automated inventory systems achieved significant improvements in workflow efficiency and notably reduced the time required to complete various inventory-related processes. According to their findings, automation eliminates or simplifies many repetitive, routine tasks

such as manual recording, counting, and data entry activities that are not only time-consuming but also prone to human fatigue and mistakes. By taking over these basic functions, the system streamlines daily operations and speeds up processing across the board. Furthermore, the study highlighted a key strategic benefit: when employees are freed from tedious, repetitive work, they can redirect their time and effort toward higher-value activities, such as analyzing inventory data, improving supplier relationships, enhancing customer service, and planning for business growth.

The study revealed that computerized systems improve resource utilization because businesses can allocate labor and materials more effectively. Managers can monitor inventory movement in real time and respond immediately to operational issues. This responsiveness improves business continuity and customer service quality.

Slack and his colleagues also emphasized the role of inventory systems in supporting strategic planning. Accurate inventory information helps organizations forecast demand, prepare budgets, and evaluate business performance. The study concluded that inventory automation contributes to organizational growth and long-term sustainability.

The researchers also highlighted that automation supports better communication between departments such as purchasing, sales, warehousing, and accounting. Because information is centralized, employees can access updated inventory data whenever necessary. This accessibility improves collaboration and decision-making within organizations.

A study conducted by Laudon and Laudon (2021) discussed the importance of management information systems in modern business operations. Their work highlighted that computerized systems improve organizational decision-making by providing accurate, timely, and accessible information. They noted that inventory management systems equipped with reporting and analytics tools enable businesses to evaluate performance and identify operational trends effectively.

Laudon and Laudon further discussed how digital systems support managerial decision-making. Managers rely on accurate inventory information to determine purchasing needs, evaluate supplier performance, and monitor business profitability. Without reliable inventory systems, businesses may encounter delays, financial losses, and operational inefficiencies.

The study also emphasized cybersecurity and data protection in computerized systems. Because inventory information is stored digitally, businesses must ensure proper security measures to prevent unauthorized access and data loss. Laudon and Laudon concluded that effective inventory systems require both technological infrastructure and organizational support.

Research conducted by Turban, Pollard, and Wood (2020) highlighted the role of enterprise systems in integrating inventory management with other business functions. Their study explained that enterprise resource planning (ERP) systems enable organizations to connect inventory, accounting, purchasing, sales, and customer service operations into a centralized platform.

The researchers found that integrated systems improve organizational coordination and reduce duplication of tasks. Employees can easily share information across departments, improving efficiency and communication. Integrated systems also provide real-time reports that support strategic planning and operational monitoring.

Turban and his colleagues further explained that ERP supported inventory systems enhance transparency and accountability. Managers can monitor inventory transactions, identify discrepancies, and evaluate business performance more effectively. The study concluded that integrated enterprise systems contribute to operational excellence and competitive advantage.

A study conducted by Zhang and Chen (2021) carried out a study focusing on intelligent inventory systems within retail businesses across various regions in Asia. They pointed out that retail operations typically involve handling thousands of product transactions every single day, a volume that makes manual methods of inventory monitoring not only highly inefficient but also unreliable and prone to frequent errors. Their research findings showed that businesses which adopted intelligent inventory systems achieved significant improvements: they attained much higher levels of stock accuracy, processed transactions at a considerably faster pace, and reduced operational delays throughout their supply and sales processes. A key advantage highlighted in their work was the use of real-time monitoring technologies, which allowed businesses to detect inventory shortages or low stock levels immediately as they occurred, enabling timely and more effective replenishment actions. The study ultimately concluded that these intelligent systems greatly enhance operational responsiveness, helping businesses adapt quickly to changes in demand, and directly contribute to higher customer satisfaction by ensuring product availability and smooth service delivery.

This study strongly supports the present research, as it validates that for businesses dealing with frequent transactions and multiple product lines such as the cleaning supply distribution business in this study implementing a digital inventory system is essential. It confirms that automation and real-time monitoring are effective solutions to the limitations of manual processes, leading to better accuracy, faster operations, and improved ability to meet customer needs.

Lee and Park (2020) conducted a study centered on the application of automated sales analysis systems in large-scale distribution enterprises. They explained that these advanced analytical technologies enable organizations to systematically uncover valuable insights, including recurring purchasing patterns, seasonal demand fluctuations, and shifting customer preferences. By processing large volumes of transaction data, these systems transform raw information into clear, actionable intelligence that would be difficult or time-consuming to gather through manual means. Their research findings showed that businesses utilizing such analytical systems achieved notably higher accuracy in demand forecasting and carried out far more effective inventory planning. Instead of relying on guesswork or general estimates, management could base their purchasing and replenishment decisions on factual data and proven trends. The study further concluded that sales analysis tools make a substantial contribution to overall profitability. This happens because the business is able to maintain optimal stock levels avoiding both shortages and overstocking align product offerings with what customers actually want, and allocate resources efficiently to areas that generate the highest returns.

This study provides strong theoretical and practical support for the current research. It confirms that the sales analysis feature included in the proposed system is a vital component, not just an added function. For the cleaning supply distribution business, this means being able to identify which products are in demand during specific periods, understand customer buying habits, and plan procurement strategically. It validates that by using data-driven insights, the business can make smarter decisions, reduce unnecessary costs, and ultimately increase its profitability and competitive advantage.

Research conducted by Wang and Li (2021) explored the implementation of cloud-based inventory systems in multinational organizations. The researchers emphasized that cloud technologies improve operational accessibility because managers and employees can access inventory information remotely through internet-based platforms. Businesses implementing cloud systems achieved better communication, real-time reporting, and more flexible operational management. The study concluded that cloud-based systems improve organizational coordination and operational efficiency.

Furthermore, the study highlighted that cloud-based systems improve operational flexibility within multinational organizations. Businesses operating across multiple branches or international locations require centralized systems that allow efficient coordination of inventory activities. Cloud technologies enable organizations to monitor inventory movement, stock levels, and sales transactions across different locations simultaneously. This capability improves inventory visibility and allows businesses to allocate resources more efficiently. Organizations implementing cloud systems achieved improved decision-making because managers could access updated operational data instantly.

Another important foreign study by Smith and Johnson (2020) examined operational challenges associated with manual inventory systems. The researchers explained that handwritten inventory records and spreadsheet-based monitoring methods are highly vulnerable to human errors and delays. Their findings revealed that organizations using manual systems frequently experienced stock discrepancies, delayed reporting, and inefficient inventory tracking. Businesses implementing computerized inventory systems reduced operational errors and improved inventory visibility.

The researchers also emphasized that manual inventory systems negatively affect operational transparency and accountability. Paper-based records are difficult to organize, retrieve, and verify, making operational monitoring more challenging. Computerized inventory systems provide electronic records and transaction histories that improve accountability and strengthen organizational control. Managers can monitor employee activities, inventory movements, and transaction records more effectively through digital systems. The study concluded that computerized technologies improve operational transparency and organizational efficiency significantly. Their research findings showed that organizations relying on manual systems commonly faced serious issues: frequent inventory discrepancies where recorded stock levels did not match actual physical counts, delayed reporting that made decision-making outdated or unreliable, and inefficient tracking that made it difficult to know exactly where items were or how much was available at any given time. In contrast, businesses that transitioned to computerized inventory systems experienced a clear improvement: operational errors were significantly reduced, and inventory visibility was greatly enhanced, allowing management to view accurate, real-time data on stock levels, movement, and status at all times.

2.2 Local Studies

Dela Cruz (2022) examined computerized inventory systems for small businesses and reported that digital inventory management can improve reporting accuracy and transaction processing. The study also highlighted the problems associated with handwritten records and manual inventory computations. This is closely related to the present study because the identified problems of the cleaning supply distributor include manual recording, inaccurate stock counts, and time-consuming

report preparation. The proposed system addresses these concerns by centralizing inventory records and automating stock and sales transactions. [5]

Santos (2021) discussed the role of system automation in improving productivity in retail and distribution operations. The study emphasizes that automation can reduce repetitive manual work and allow employees to spend more time on other operational activities. This is relevant to the proposed system because one of its purposes is to reduce manual inventory and sales recording. By providing automated transaction recording and report generation, the proposed system aims to make daily inventory and sales activities more organized and efficient. [6]

2.3 Related Literature

Inventory management is an essential component of business operations because it involves monitoring product quantities, recording stock movements, and maintaining sufficient stock levels to support daily activities. In distribution businesses, accurate inventory information is necessary to determine the availability of products and identify when replenishment is required. Stevenson explains that effective inventory management supports operational efficiency by helping organizations control stock levels, manage demand, and coordinate supply activities [1].

Traditional inventory processes commonly rely on handwritten records or separate spreadsheet files, which may result in inaccurate quantities, delayed updates, and difficulties in retrieving historical information. Computerized inventory systems provide a more organized approach by recording transactions digitally and maintaining centralized product information. This supports the proposed system, which is designed to replace manual inventory recording with a centralized platform for managing cleaning supply products.

2.4.2 Real-Time Inventory Visibility and Supply Chain Information

Real-time inventory visibility allows businesses to obtain updated information regarding product availability, stock movement, and inventory levels. Chopra and Meindl emphasize the importance of accurate and timely information in supply-chain management because information enables organizations to coordinate activities and respond to changes in demand [2]. When inventory information is updated consistently, businesses can better monitor available stock and identify products that require replenishment.

For a cleaning supply distribution business, real-time inventory information is particularly important because products are continuously received, released, and sold. A centralized information system can record these activities and update the corresponding product quantities. The proposed system applies this concept by recording stock-in and stock-out transactions and providing updated inventory information through the system dashboard.

2.4.3 Sales Analytics and Product Movement Analysis

Sales analytics provides organizations with information that can be used to understand product performance, sales trends, and customer demand. The integration of sales information with inventory management allows businesses to identify products that move frequently and products that remain in stock for longer periods. Gupta and Kohli emphasize the importance of integrating analytical information with business operations to support better planning and decision-making [3].

The identification of fast-moving and slow-moving products is relevant to inventory planning because it provides information about product demand. The proposed Intelligent Stock and Sales Analysis System incorporates sales analysis functions that use recorded sales information to identify product movement patterns. These results can assist the user in monitoring product performance and making informed inventory-related decisions.

2.4.4 Inventory Accuracy and Transaction Monitoring

Inventory accuracy is important in distribution operations because incorrect stock information can affect purchasing, sales, order fulfillment, and customer service. Piasecki emphasizes that maintaining accurate inventory records is necessary for

effective warehouse and inventory control [4]. Errors in recorded quantities may result in shortages, excess stock, incorrect orders, and additional operational costs.

Digital transaction recording can help improve inventory accuracy by providing a systematic record of stock movements. Instead of relying entirely on manual updates, the proposed system records stock-in and stock-out transactions and uses these records to update product quantities. The system also maintains transaction information that can be reviewed when monitoring inventory activities.

2.4.5 Supply Chain Management and Information Technology

Information technology has become an important component of modern supply-chain management because it allows businesses to coordinate information related to products, transactions, suppliers, and customers. Christopher explains that effective supply-chain management depends on the coordination and visibility of information across business activities [5]. Digital systems can therefore support organizations in improving information flow and reducing delays in operational processes.

For small and medium-sized distribution businesses, centralized information management can provide a structured way to organize inventory and sales activities. The proposed system applies this concept by combining product management, inventory monitoring, sales recording, reporting, and sales analysis into a single platform designed for cleaning supply distribution.

2.4.6 Automation and Computerized Business Operations

Automation can reduce repetitive manual activities and improve the consistency of business processes. Heizer, Render, and Munson discuss the role of technology in operations management, including inventory management, forecasting, and supply-chain activities [6]. Computerized systems can help organizations process information more efficiently while reducing dependence on manual recordkeeping.

Local studies included in the original Chapter 2 also describe the benefits of computerized systems for small businesses. Dela Cruz reported that digital inventory systems can improve transaction processing and reporting while reducing errors associated with manual records [7]. Santos similarly emphasized that automation can improve employee productivity and operational efficiency by reducing repetitive tasks [8]. These findings support the development of the proposed system for automating inventory and sales-related activities.

2.4.7 Database Management and Centralized Information Systems

A centralized database provides a structured environment for storing, organizing, retrieving, and updating business information. Database-driven systems allow related records to be connected and accessed by different system modules, reducing the need to maintain separate records. This approach is particularly useful for inventory and sales systems because product, transaction, and sales information must remain consistent across different operations.

The proposed system uses a centralized database to manage product information, inventory logs, transactions, sales records, and analytical information. The centralized structure allows inventory transactions to be recorded and used by reporting and sales analysis functions, providing an integrated source of information for the system.

2.4.8 Software Engineering and System Development

Software engineering provides structured approaches for analyzing requirements, designing systems, implementing functions, testing applications, and maintaining software. Pressman and Maxim describe software development as a systematic process involving activities that support the development of reliable and maintainable software [9].

The proposed Intelligent Stock and Sales Analysis System applies these principles through the Agile development methodology. The system is developed through iterative activities that include requirements gathering, planning, system

design, development, testing, deployment, and maintenance. This approach allows system functions to be evaluated and improved throughout the development process.

2.4.9 Intelligent Inventory Systems and Data-Driven Decision-Making

The development of intelligent inventory systems has expanded the role of information systems beyond simple record keeping. Modern inventory applications can combine historical transaction data, sales information, and analytical functions to provide information about product demand and inventory movement. Such systems support data-driven decision-making by transforming stored operational data into useful business information.

The proposed system incorporates this concept by combining inventory management with sales analysis. Instead of only recording product quantities, the system analyzes sales and inventory information to identify product movement patterns and generate reports. This integration provides the business with organized information that can support inventory monitoring, sales analysis, and future planning.

2.4.10 Research Gap

The reviewed literature establishes the importance of inventory management, real-time information, sales analysis, inventory accuracy, automation, centralized databases, and software-based business operations. However, much of the existing literature focuses on general inventory management, large organizations, retail operations, manufacturing, and broader supply-chain environments.

The proposed study addresses this gap by developing an Intelligent Stock and Sales Analysis System specifically for a cleaning supply distribution business. The system combines product management, stock monitoring, stock-in and stock-out recording, sales transactions, reporting, and sales analysis in one centralized platform. This provides an application of the concepts discussed in the reviewed literature to the specific operational requirements of the target business.

2.4 Synthesis of the Reviewed Literature and Studies

The reviewed foreign and local literature shows that manual inventory processes can create inaccurate records, delayed reporting, and difficulties in monitoring stock. Computerized systems provide a way to centralize information, improve transaction recording, and make inventory information easier to access. The reviewed studies also support the use of sales analytics to identify fast-moving and slow-moving products and assist in inventory-related decisions.

However, much of the reviewed literature focuses on general inventory management, retail, manufacturing, or larger distribution environments. The present study addresses a more specific context: a small-scale cleaning supply distribution business. The proposed Intelligent Stock and Sales Analysis System combines product management, stock-in and stock-out recording, inventory monitoring, sales recording, sales analysis, and report generation in one system. This provides the research gap addressed by the study.

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CHAPTER III

METHODOLOGY

3.1 System Architecture

The system uses a centralized web-based client-server architecture. The presentation layer provides the user interface, the application layer processes inventory and sales operations, and the data layer stores product, transaction, sales, user, and analytical records. This structure supports consistent information management and access to current operational data [2].

The Presentation Layer contains the login page, dashboard, product management, inventory monitoring, stock-in and stock-out management, sales transaction management, sales analytics dashboard, report generation, and user account management. The Application Layer handles authentication, inventory transactions, sales records, analysis, dashboards, and reports. The Data Layer stores user accounts, product information, product categories, inventory records, stock transactions, sales transactions, customer records, analytics data, and generated reports.

- **Presentation Layer:** Provides the web-based interface through which the administrator accesses the dashboard, product management, inventory monitoring, stock-in and stock-out transactions, sales records, reports, and sales analysis. The interface is designed to provide clear access to the system's major functions.

- **Application Layer:** Handles the system's core business processes, including user authentication, product management, stock transaction processing, sales recording, inventory updates, report generation, and sales analysis. This layer processes user requests and applies the rules required for each transaction.

- **Data Layer:** Uses the database to store product information, user accounts, stock transactions, sales records, inventory logs, audit information, dispatch records, and analytical results. The centralized data layer allows the system to maintain consistent records and retrieve information for reporting and analysis.

3.2 Hardware and Software Environment

The proposed system requires appropriate hardware and software resources to support inventory management, sales transactions, real-time inventory monitoring, sales analysis, and report generation. The selected resources provide the necessary environment for developing and operating the system.

Table 4.1 presents the recommended hardware specifications.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher

Storage	256 GB SSD	512 GB SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provides sufficient processing capability for executing analytical operations and rendering graphical dashboards without significant delays.

Table 4.2 presents the software requirements.

Windows 11 / Ubuntu Linux	Operating System
Visual Studio Code	Source Code Editor
MySQL	Database Management System
PHP / Node.js	Backend Development
ReactJS	Frontend Development
Google Chrome	Web Browser
Git	Version Control
XAMPP	Local Development Server

The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

3.3 Data Gathering Techniques

To identify the actual requirements of the cleaning supply distribution business, the researchers gathered information concerning the existing inventory and sales processes. The collected information was used to determine the problems encountered in manual stock monitoring and to identify the functions required in the proposed system.

Interview: Information was gathered from the business context to understand the existing procedures for recording products, monitoring stock, recording stock-in and stock-out transactions, and preparing sales information. The interview information helped identify the requirements that needed to be addressed by the proposed system.

Observation: The researchers examined the existing workflow to understand how inventory and sales records are maintained and how stock information is updated. This helped identify areas where manual processes could result in delays, inaccurate records, or difficulty monitoring product movement.

Document Review: Existing records and business documents were reviewed to determine the information required for product, inventory, and sales records. The information obtained from these documents was used as a basis for designing the database and system functions.

3.4 Testing Procedures

To guarantee structural integrity and prevent database corruption or logical errors during live transaction flows, the system will undergoes a progressive, three-tier testing framework. This structured methodology moves from isolated backend code modules to complete system-wide integrations and end-user field evaluations:

- **Unit Testing**: Individual functions and modules are tested separately. This includes product management, stock-in and stock-out processing, sales recording, inventory updates, and report-generation functions.

- **System Testing**: The complete system is tested as an integrated application. The researchers verify whether product information, inventory transactions, sales records, reports, and analytical functions work together correctly and whether database operations maintain accurate information.

- **User Acceptance Testing (UAT)**: The system is evaluated by intended users to determine whether it is understandable, usable, and suitable for the intended business processes. Feedback from the evaluation is used to identify possible improvements.

3.5 System Development Methodology

The Agile process used in the study consists of requirements analysis, system planning and design, development, testing, deployment, and maintenance. Iterative development allows functions to be reviewed and improved as the system progresses. This approach is consistent with software-engineering discussions of agility and process [3].

Phase	**Activities**
Requirements	Identify current problems, users, requirements, and system scope.
Design	Prepare architecture, workflow, database structure, and interface design.
Development	Implement product, stock, sales, reporting, and analysis functions.
Testing	Test modules, integrated functions, usability, security, accuracy, and efficiency.
Deployment	Configure the application and database and prepare the system for use.
Maintenance	Correct issues, update functions, and improve the system based on feedback.

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CHAPTER IV

RESULTS AND DISCUSSION

4.1 Overview of the Developed System

The developed **Intelligent Stock and Sales Analysis System** provides a centralized platform for managing product information, inventory transactions, sales records, reports, and sales analysis. The system records stock-in and stock-out transactions, updates product quantities, records sales transactions, and organizes stored information for inventory monitoring and reporting.

The system supports the monitoring of product movement by analyzing recorded sales and inventory transactions. It provides information that can be used to identify **fast-moving and slow-moving products**, allowing the user to examine product performance and inventory movement. The system also provides a dashboard that presents summarized inventory and sales information for easier monitoring of current operations.

The inventory module manages product information and stock quantities. Stock-in transactions are recorded when products are received, while stock-out transactions are recorded when products are released or sold. These transactions provide a record of inventory movement and allow the system to maintain updated product quantities.

The sales module records sales transactions and stores the corresponding product and transaction information. The recorded sales data are used by the sales analysis functions to identify product movement patterns and provide information regarding sales performance. This information can also support inventory monitoring and planning.

The reporting module organizes stored inventory and sales information into reports. These reports include sales information, stock movement, inventory status, and product performance. The centralized records allow the user to retrieve and review information without relying solely on separate manual records.

The system integrates **product management, inventory monitoring, stock-in and stock-out transactions, sales recording, reporting, and sales analysis** into a single platform. This integration addresses the problems identified in the existing process, including manual recording, inaccurate stock information, difficulty monitoring product movement, and time-consuming report preparation.

Overall, the developed system provides an organized and centralized approach to managing inventory and sales information for the cleaning supply distribution business. Its integrated functions support the monitoring of stock and sales activities and provide organized information for operational reporting and analysis.

The developed system was created with the following objectives:

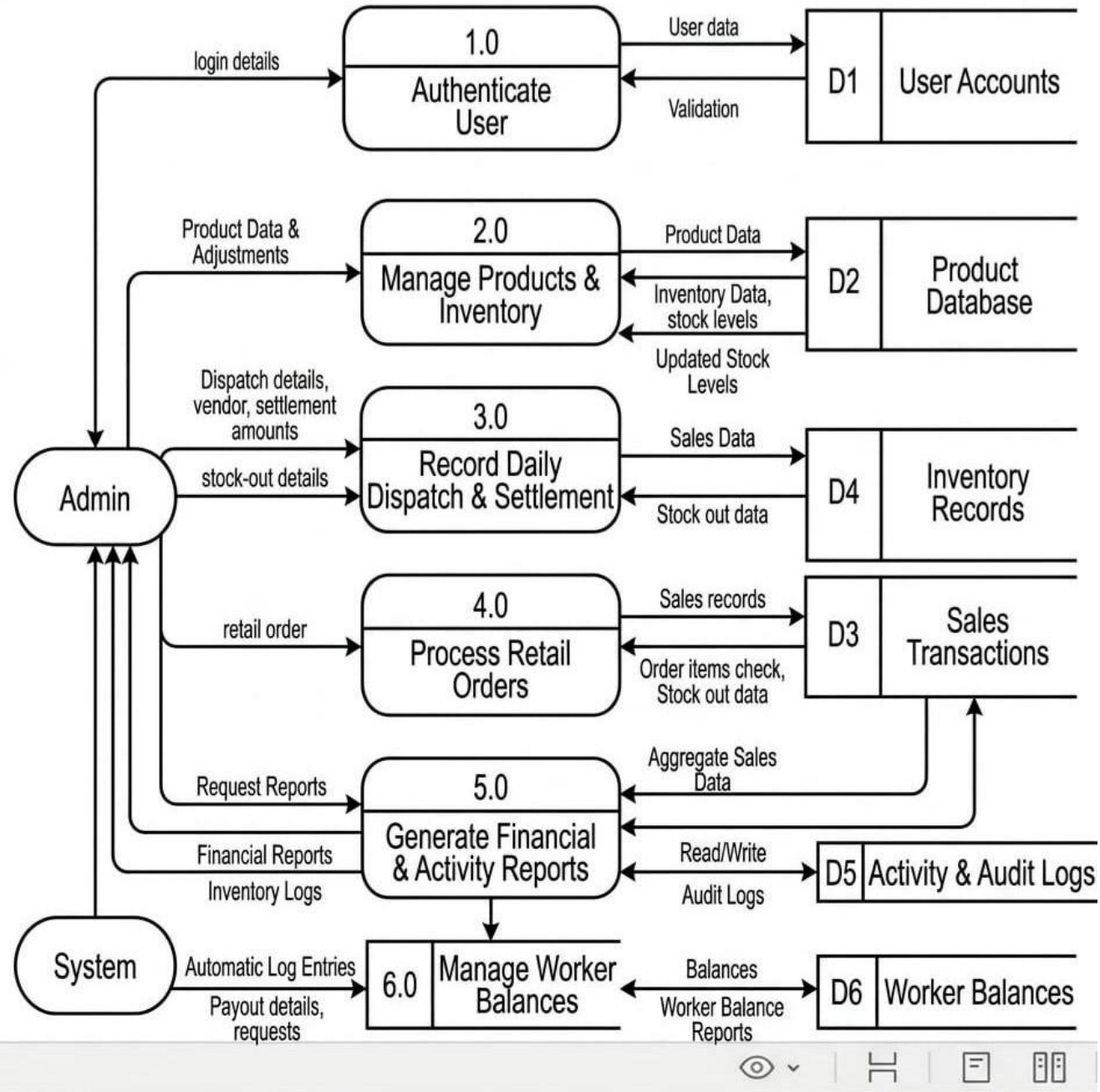
- To analyze the current inventory process of the cleaning supply distributor.
- To design a user-friendly inventory and sales management interface.
- To develop a system for recording stock-in and stock-out transactions.
- To implement real-time inventory monitoring.
- To integrate sales analysis features to identify product movement trends.

- To generate automated inventory and sales reports.
- To test and evaluate the system's functionality, usability, and performance.

The development of the **Intelligent Stock and Sales Analysis System** followed the Agile software development methodology. It incorporated a web-based user interface, user authentication, centralized database management, inventory transaction processing, sales recording, reporting, and sales analysis. These features were designed to make daily inventory and sales operations more organized, accurate, and efficient.

The completed system consisted of several integrated modules that work together to support the different functions of the cleaning supply distribution business. These modules include user authentication, product management, inventory monitoring, stock-in and stock-out transactions, sales management, reporting, and sales analysis. Since these modules use centralized records, inventory and sales information can be organized and accessed throughout the system.

Figure 4.1 presents the overall workflow of the developed system.



The workflow begins with **user authentication through login**, allowing the authorized user to access the system dashboard and its available functions. After logging in, the user can manage product information, monitor inventory levels, record stock-in and stock-out transactions, and manage sales records. When new products are received, the user records the stock-in transaction, and the system automatically updates the corresponding product quantity. For sales or product releases, the user records the stock-out transaction, which also updates the available inventory.

The system records each transaction in the centralized database, allowing the user to monitor product quantities and review stock movement history. Sales transactions are also recorded and organized according to the products sold, quantities, dates, and total sales amounts. The system uses these records to identify product movement patterns, including fast-moving and slow-moving products, and provides information that can support inventory monitoring and restocking decisions.

The **Reports and Sales Analysis** module processes the recorded inventory and sales information and generates organized reports on sales performance, stock movement, inventory status, and product performance. The dashboard presents summarized information such as available products, low-stock items, sales information, and other relevant indicators. Finally, all transactions and generated information are stored in the centralized database, allowing the user to monitor business operations and support data-driven decision-making.

4.2 System Development Methodology

The development of the **Intelligent Stock and Sales Analysis System** followed the **Agile Software Development Methodology**, utilizing an iterative approach that allowed continuous improvement, feedback, and adaptation to the operational requirements of the cleaning supply distribution business across seven distinct phases:

- **Requirements Gathering:** Information was collected through interviews, observations, and review of existing inventory and sales records to identify operational problems such as manual stock recording, inaccurate inventory counts, delayed reporting, and difficulty monitoring product movement. These findings were used to establish the functional and non-functional requirements of the system.

- **System Planning:** The researchers defined the project scope, objectives, development schedule, required hardware and software resources, and major system functions. The planning phase established the activities and resources needed for the structured development of the proposed system.

- **System Design:** The researchers developed the system design using the **Data Flow Diagram (DFD), Entity Relationship Diagram (ERD), Use Case Diagram, system architecture, database structure, and interface designs**. These designs established the system workflow, relationships among data, user interactions, and overall structure of the application.

- **Software Development:** The design specifications were translated into a functional web-based system. The development focused on implementing the major functions of the system, including user authentication, product management, inventory monitoring, stock-in and stock-out transactions, sales recording, reporting, and sales analysis. The functions were developed incrementally to identify and address errors during the development process.

- **Testing:** The developed system underwent testing to verify that its functions operated correctly and according to the identified requirements. Testing included checking individual functions, integrated system operations, database transactions, inventory updates, sales processing, report generation, and user interaction. The system was also evaluated based on functionality, usability, accuracy, security, and efficiency.

- **Deployment:** After testing and necessary revisions, the completed system was prepared for deployment in the target environment. The database and system components were configured, and the developed functions were validated using sample operational data to ensure that the system could properly process inventory and sales transactions.

- **Maintenance:** Post-development maintenance involves monitoring system performance, correcting identified errors, updating system functions, and improving the application based on user feedback. Future enhancements may include additional analytical functions, automated notification.

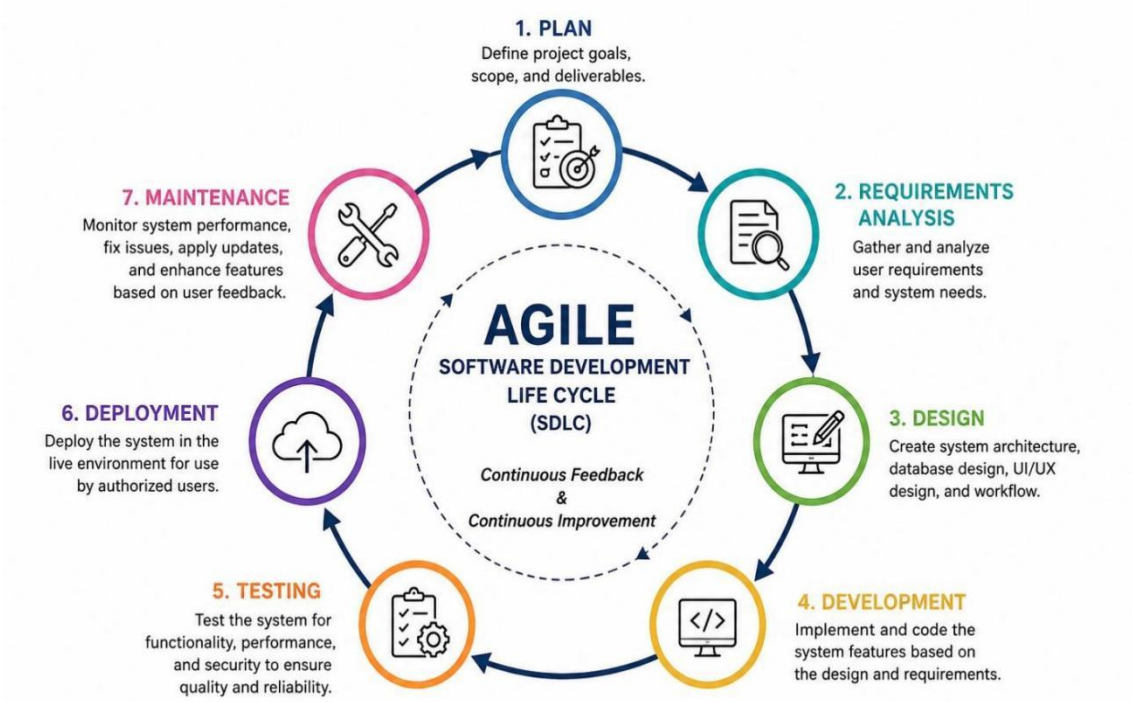


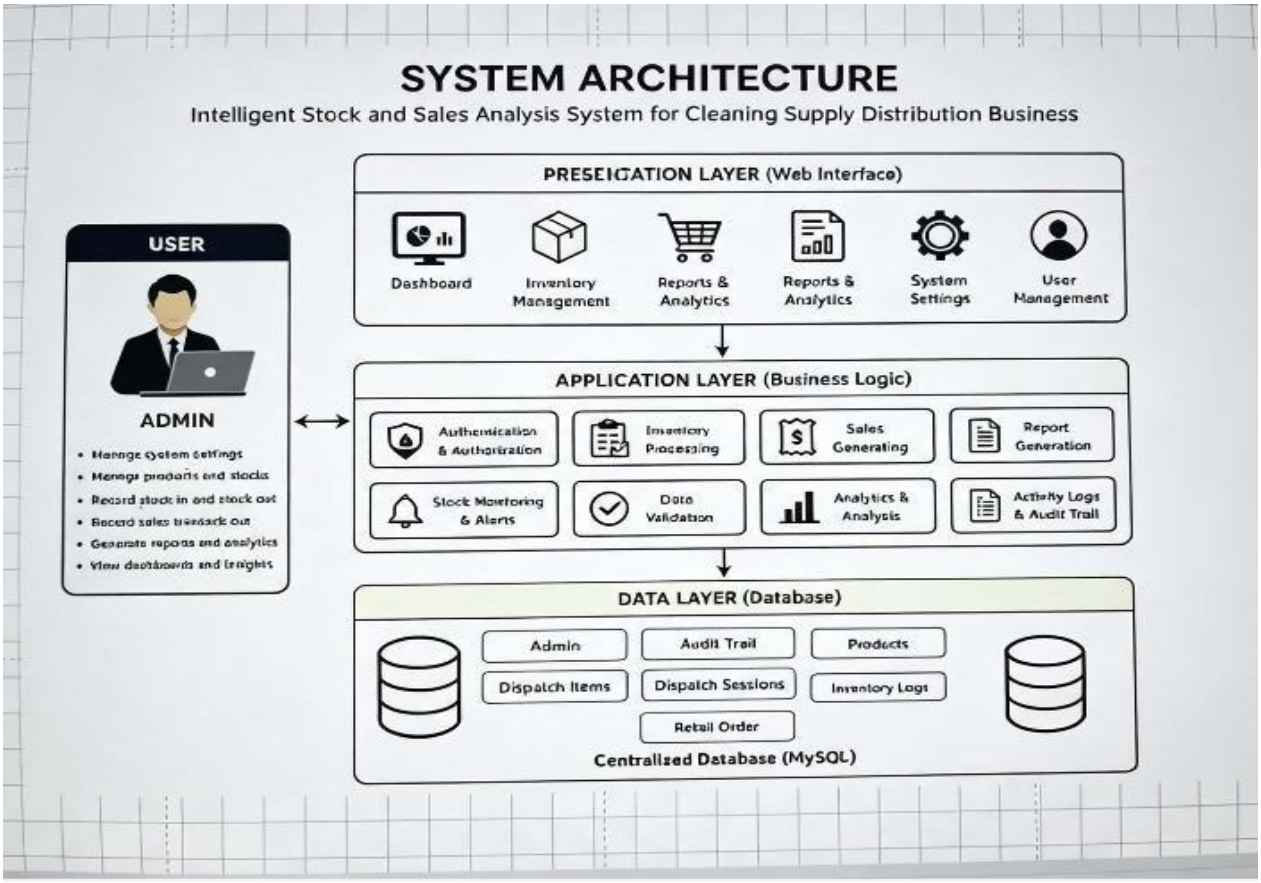
Figure 4.2. Agile Software Development Life Cycle Model

4.3 System Architecture

The system uses a centralized web-based client-server architecture. The presentation layer provides the user interface, the application layer processes inventory and sales operations, and the data layer stores product, transaction, sales, user, and analytical records. This structure supports consistent information management and access to current operational data [2].

The Presentation Layer contains the login page, dashboard, product management, inventory monitoring, stock-in and stock-out management, sales transaction management, sales analytics dashboard, report generation, and user account management. The Application Layer handles authentication, inventory transactions, sales records, analysis, dashboards, and reports. The Data Layer stores user accounts, product information, product categories, inventory records, stock transactions, sales transactions, customer records, analytics data, and generated reports.

Figure 4.3 illustrates the overall system architecture of the developed **Intelligent Stock & Sales Analysis**



The **Intelligent Stock and Sales Analysis System** is built on a **Three-Tier Client-Server Architecture** that isolates the user interface, core business logic, and database management into distinct, interconnected operational layers

1. Presentation Layer (Front-End Interface)

- **Role:** Acts as the primary user-facing layer through which the authorized user accesses the system using a web browser through a local or network connection.
- **Key Components:** Provides accessible and organized modules including the **Login Page, Dashboard, Product Management, Inventory Monitoring, Stock-In Management, Stock-Out Management, Sales Management, Reports, and Sales Analysis**.

2. Application Layer (Core Business Logic)

- **Role:** Serves as the communication bridge between the user interface and the database, handling user authentication, data validation, transaction processing, and the business rules required by the system.
- **Core Operations:** Manages product processing, stock-in and stock-out transactions, inventory quantity updates, sales recording, low-stock monitoring, sales analysis, report generation, and dashboard information before storing or retrieving data from the database.

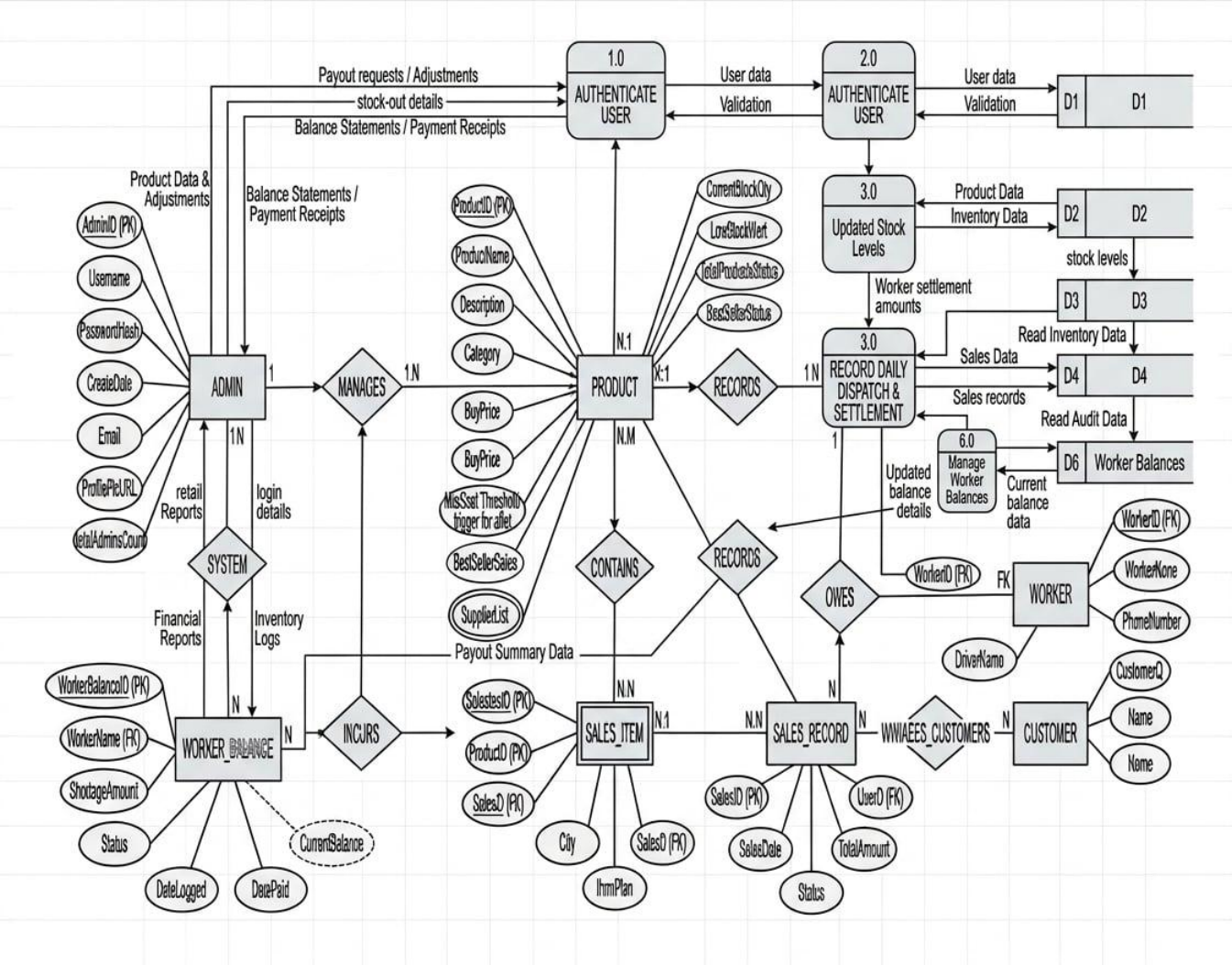
3. Data Layer (Central Repository)

- **Role:** Serves as the centralized data repository that stores, organizes, and retrieves the information required by the system. The database uses structured relationships between records to maintain consistent and organized information.

- **Stored Datasets:** Contains **Admin, Audit Trail, Dispatch Items, Dispatch Sessions, Inventory Logs, Products, Retail Orders, and Machine Learning** data required for inventory monitoring, transaction processing, reporting, and analysis.
- **Data Integration:** Synchronizes information across system modules by updating inventory quantities after stock-in and stock-out transactions, recording sales information, maintaining transaction logs, and providing updated data for reports and sales analysis.

4.4 Database Design

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The database of the Intelligent Stock and Sales Analysis System utilizes a centralized relational database to store and manage information required for inventory, sales, product, customer, transaction, and analytical operations. The database structure uses primary and foreign keys to establish relationships between related records, maintain data consistency, and support organized data retrieval. This design allows the different system modules to access and update the required information for inventory monitoring, sales processing, reporting, and sales analysis.

The primary entities of the database include the following:

- Admin
- Products
- Categories
- Customers
- Transactions
- Transaction Details
- Inventory Logs
- Dispatch Sessions
- Dispatch Items
- Retail Orders
- Audit Trail
- Sales Analytics
- Analytics Results
- Machine Learning

Admin Table

This table stores account credentials and user information

Field	Description
ID	Primary Key
Full Name	Full Display name of admin
Username	Unique handle used by the user to login
Password	The securely hashed password
Email	Recovery Email for one side login for admin

Products Table

This table stores product information

Field	Description
Id	Primary Key
Category	Grouping or classification name
Product	Name The official name of the product
Variation	Specification of product details
Description	Detailed Text that describes the products
Wholesales_Price	Discount Cost Per Unit For Wholesale purchase
Retail_Price	Standard selling price per unit for individual customer
Quantity	Current available stock/inventory count on hand
Image	Referred to product what look like
Created_at	A timestamp recorded when the product entry was created

Sales Analytics Table

The Sales Analytics table stores analytical information generated by the system based on inventory and sales transactions. This table contains summarized data that helps business owners monitor sales performance, identify product movement trends, and support inventory planning and decision-making.

Field	Description
analytics_id	Primary Key
product_id	Foreign Key
total_quantity_sold	Total quantity of the product sold
total_sales	Total sales amount generated by the product
product_status	Indicates whether the product is fast-moving or slow-moving
analysis_date	Date the sales analysis was generated

The relationships among these tables enable the system to monitor inventory movement, analyze sales performance, identify fast-moving and slow-moving products, and generate comprehensive inventory and sales reports. Proper indexing and foreign key constraints help maintain data integrity, improve query performance, and ensure that the system efficiently supports both daily business transactions and sales analysis.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary, conclusion, impact, and recommendations of the Intelligent Stock and Sales Analysis System for a Cleaning Supply Distribution Business. It highlights how the developed system addresses the operational problems identified in the earlier chapters.The study was conducted to address the limitations of traditional and manual inventory management practices commonly encountered in cleaning supply distribution businesses. Manual recording of inventory and sales transactions often results in inaccurate records, delayed updates, difficulties in monitoring stock levels, and challenges in generating reports for decision-making. These issues may lead to stock shortages, overstocking, loss of sales opportunities, and inefficient business operations. To overcome these problems, the researchers developed an Intelligent Inventory and Sales Analysis System that automates inventory management processes and provides analytical tools to support business decisions.

5.2 Summary of Findings

- The system successfully automated inventory management and sales recording, reducing manual data entry and improving the organization of stock information.
- The system provided inventory monitoring and sales tracking through an interactive dashboard.
- Comprehensive reports, including inventory, sales, stock movement, reorder, and product-performance information, were generated.
- The system improved the organization of inventory and sales information and reduced the time required for manual record preparation.
- User Acceptance Testing indicated that the system was functional, user-friendly, reliable, and efficient within the study's evaluation conditions.

5.3 Conclusion

The Intelligent Stock and Sales Analysis System has been developed as a centralized solution for inventory and sales management in a cleaning supply distribution business. It addresses manual record-keeping concerns by organizing product records, stock movements, sales transactions, reports, and sales analysis. The documented results support the importance of accurate inventory information for operational control [1] and current information for replenishment decisions [2].

The system is a functional academic prototype within the scope and testing conditions of the study. It is not intended to replace full enterprise accounting, multi-branch operations, or other functions outside the defined scope.

5.4 Recommendations

- Integrate barcode scanning to improve inventory transaction speed and accuracy.
- Implement automated low-stock notifications and reorder alerts.
- Improve report export options, including PDF, Excel, and CSV formats.
- Explore predictive analytics for inventory demand and sales trends.
- Conduct longer-term evaluation with a broader group of users and operational scenarios.
- Investigate a mobile application version for improved accessibility.
- Provide user training, regular maintenance, and database backups.

5.5 Impact of the Study

The project demonstrates how an information system can support inventory and sales operations in a small distribution business. By organizing transaction information and making stock and sales data easier to access, the system provides a foundation for more structured operational decision-making. The study also provides an academic reference for future work involving inventory management, sales analytics, and business information systems.Through the analysis of sales data and product movement trends, the system provides valuable information that helps business owners identify high-performing and low performing products, monitor sales patterns, and make informed decisions regarding inventory replenishment and stock management. The availability of accurate and timely information contributes to better planning, resource allocation, and overall business performance. As a result, businesses are able to improve productivity, reduce unnecessary costs, and maintain appropriate inventory levels to meet customer demands.

5.6 Summary and Future Work

The study achieved its objectives by analyzing the existing inventory process, developing a centralized system for product and stock management, recording sales transactions, providing sales analysis, and generating reports. Future development may extend the prototype with barcode integration, automated notifications, predictive analytics, mobile accessibility, cloud deployment, additional security controls, and broader scalability. These extensions should be evaluated separately because they are outside the documented scope of the current study.The developed system integrates inventory management and sales analysis into a centralized platform that enables users to perform daily business operations more efficiently. Through features such as product management, stock-in and stock-out recording, sales transaction processing, real-time inventory monitoring, and automated report generation, the system provides accurate and organized information that supports business decision-making. The sales analysis component also allows business owners to identify product movement trends and evaluate sales performance, helping them make informed decisions regarding inventory planning and stock replenishment.

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